



Introduction

- **Task:** Drum audio style (semantic labels for groups of drum patterns) classification comparing a CNN with a pretrained transformer based model in a Supervised Transfer Learning context.
- **Gap:** Drum audio style is under-explored in comparison to symbolic representations or single drum instrument classification.
- **Research Question:** What is the effect of pretrained general audio embeddings in PaSST when compared to a baseline CNN for drum audio style classification?

Dataset, Preprocessing, Evaluation

- **Dataset:** Groove MIDI Dataset (GMD) : 18,264 two-measure clips at 16 kHz mono with 74 combined style classes.
- **Input features:** Mel spectrograms $n_{\text{mels}} = 128$, $\text{NFFT} = 1024$.
- **Evaluation:** F1 (primary accuracy metric) also computed per-class metrics via confusion matrices.

Models, Experiments

- **CNN:** 2D conv stack, ReLU+pool, FC layer.
- **PaSST:** Patchout Audio Spectrogram Transformer pretrained on AudioSet with frozen embeddings.
- **Experiments:** 34 experiments in four categories: Configuration Exploration, Dataset Exploration, Model Depth and Patchout, Augmentations and Padding.

Limits, Future Research

- **Limits:** Dataset Class imbalance (Rock-heavy), No use of MIDI files, Limited access to compute resources.
- **Future Research:** Expand to E-GMD Dataset, Fine-tuning PaSST, Ensemble models, Unsupervised contexts.

Interpretability, Analysis

- **PaSST:** Clusters more aligned with primary style rather than individual style-class.
- **CNN:** Smaller clusters around style classes less by primary style.
- **Centroid shift:** 10x less mean centroid shift in PaSST clusters compared to CNN.
- Both confuse musically similar styles and although more accurate the CNN baseline is sensitive to class imbalance, augmentation and padding mode changes while PaSST displays a more robust and hierarchical feature representation.

Results

ID	Model	F1	Description
1.1	CNN	0.9204	Defaults with rock styles
2.2	PaSST	0.8699	PaSST config
3.2	PaSST	0.3911	GMD-mini
4.1	CNN	0.7531	GMD-full
5.4	CNN	0.7827	GMD-full repeat
6.1	PaSST	0.7247	t-SNE & Time patchout
7.1	PaSST	0.8582	PaSST MLP 3 Layers
8.3	PaSST	0.8659	PaSST MLP 4 Layers
9.2	CNN	0.8944	CNN 7 Conv Layers
10.1	CNN	0.9080	CNN 7 Conv Layers + Noise, RoomSim
11.2	PaSST	0.8752	PaSST MLP 4L, Reflection padding

Table 1. F1 scores for the best model per round.

Figure 1. PaSST t-SNE by primary style.

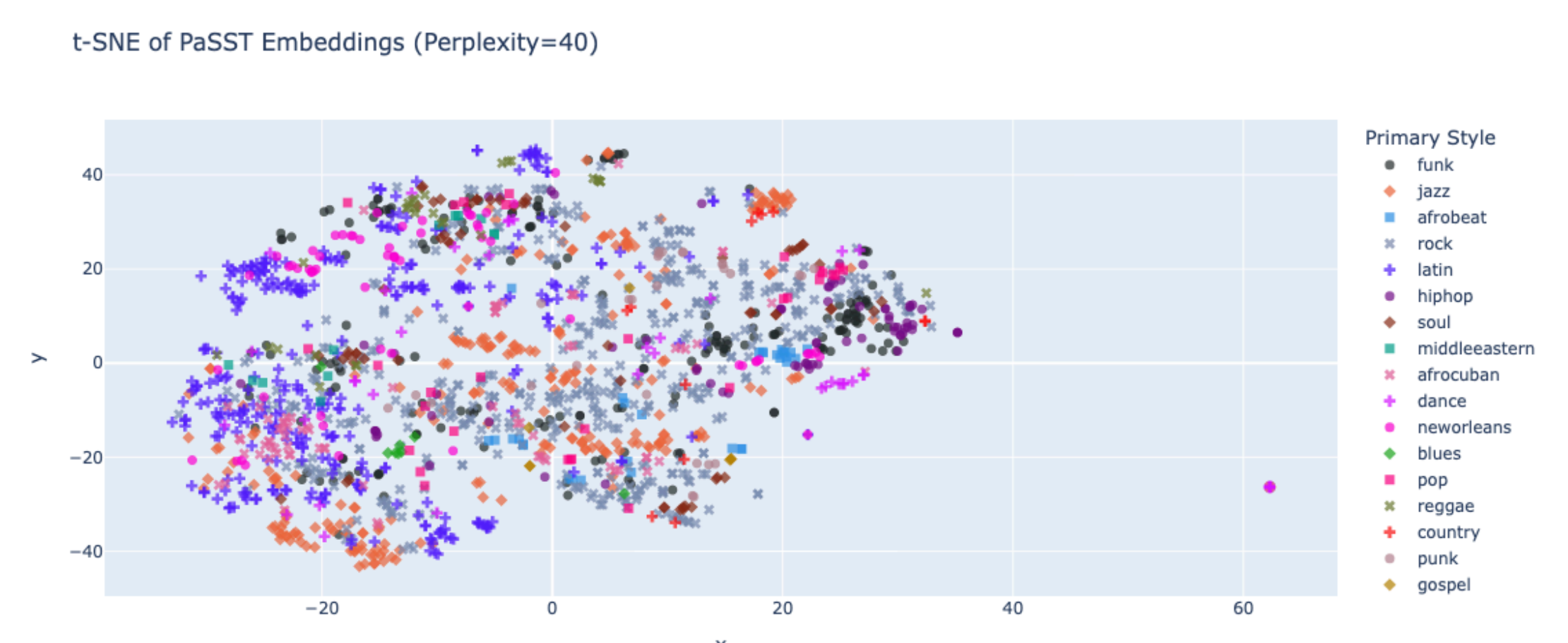
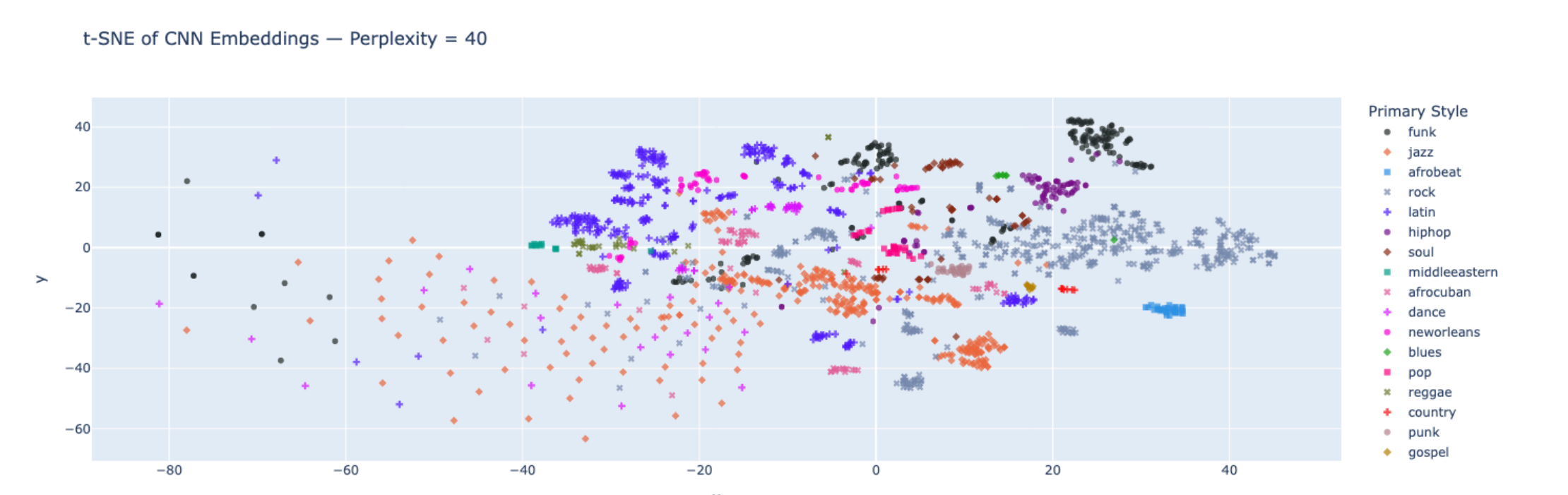
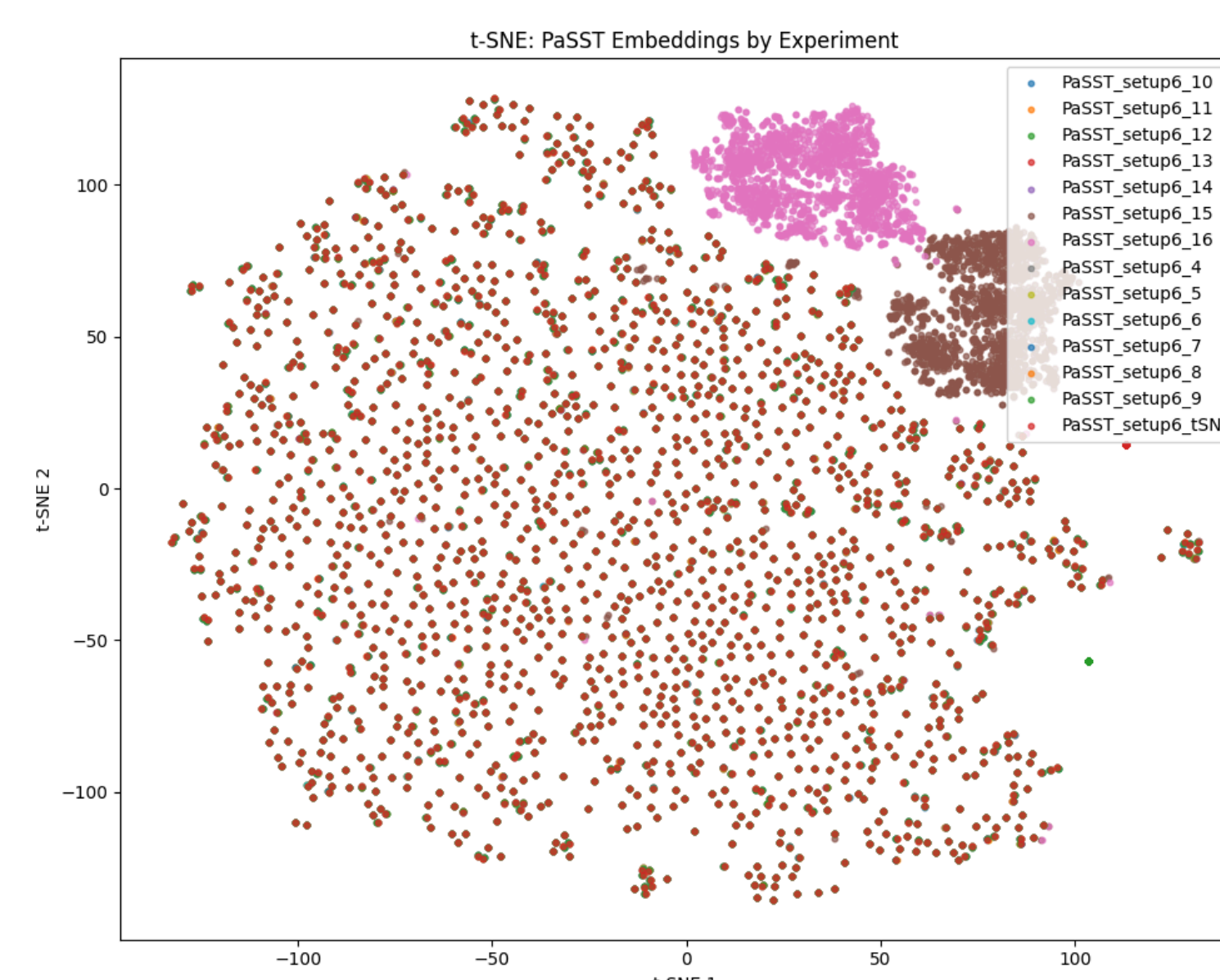


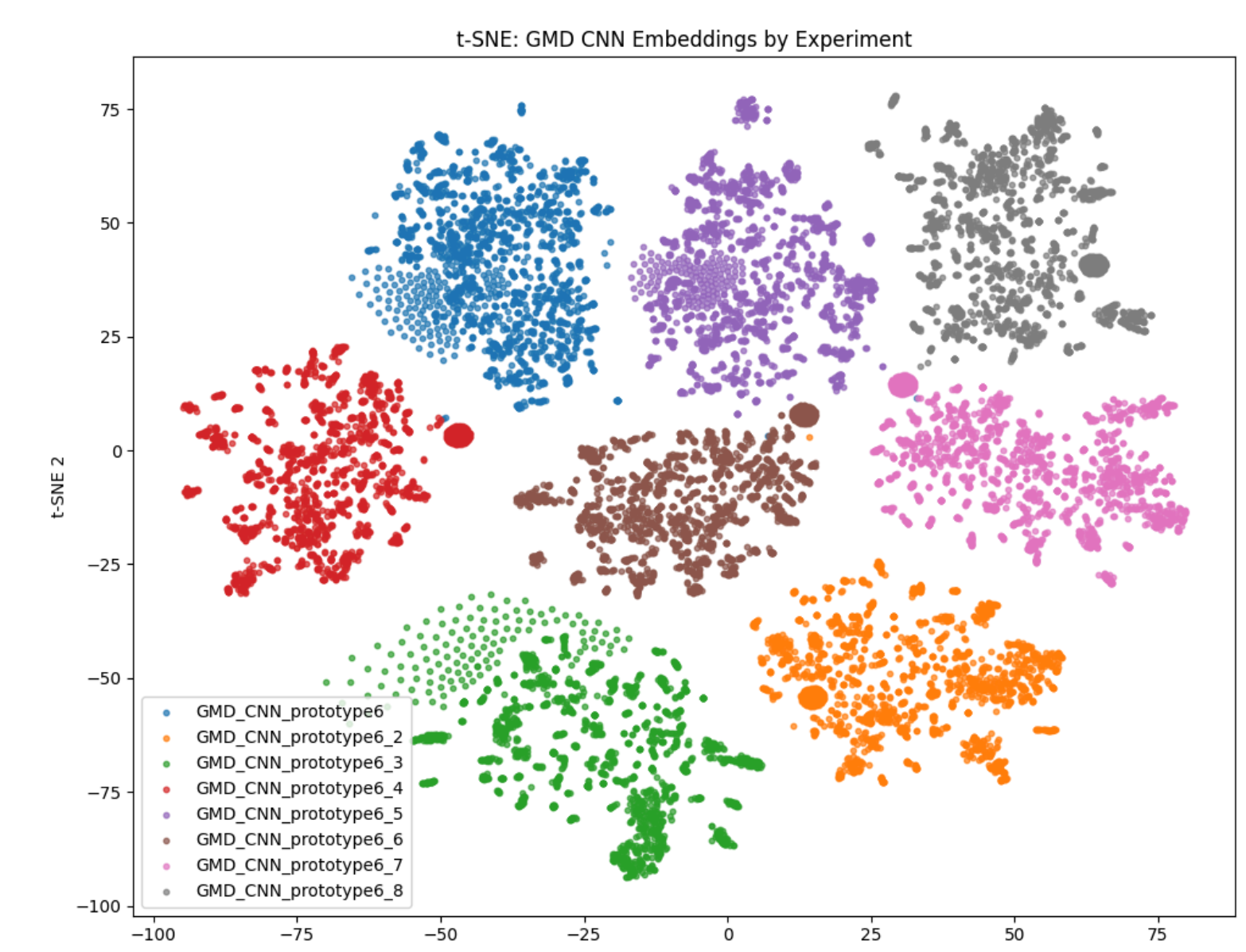
Figure 2. CNN t-SNE by primary style.



t-SNE across PaSST experiments.



t-SNE across CNN experiments.



Paper

